

Customer Churn Analysis & Insights

1. Churn Rate Overview

- The dataset indicates that **26.5% of customers have churned**, meaning more than **one-fourth** of the customer base is lost.
- Month-to-month contract holders account for **73% of total churned customers**—making them the most vulnerable group.

Key Observations:

- ✓ **Senior Citizens** have a churn rate of **41%**, compared to **23%** for non-senior citizens.
- ✓ Customers using **electronic checks** as payment method account for **45% of total churners**, much higher than those using credit cards or bank transfers.
- ✓ Customers on **two-year contracts** have the lowest churn rate of just **3%**.

Implications:

Companies should introduce incentives for month-to-month customers, provide better customer support for senior citizens, and optimize payment options.

2. Service Usage Patterns

- Customers subscribed to **Fiber-optic internet** have the highest churn rate at **40%**, while DSL users churn at only **20%**.
- Customers **without Online Security or Tech Support** churn at a rate of **38%**, whereas those with add-on services churn at just **19%**.

Key Observations:

- ✓ Customers **without Tech Support** have a churn rate of **37%**, while those with support churn at only **16%**.
- ✓ **Streaming Services (TV & Movies)** have **no major impact** on churn, suggesting entertainment options do not influence customer decisions as much.

Implications:

Improving **fiber-optic internet service quality** and **promoting add-on security/support features** can significantly enhance retention rates.

3. Payment & Charges Impact

- Customers with **monthly charges above \$70** have a churn rate of **42%**, compared to only **8%** for those with charges below \$40.

- Customers **with automatic payments** have a **15% churn rate**, much lower than **43%** for manual payers.

Key Observations:

- ✓ **High spenders churn more frequently**, possibly due to dissatisfaction with costs.
- ✓ **Customers with one-year contracts** have an intermediate churn rate of **12%**, suggesting that longer commitments reduce churn.

Implications:

Introducing **tiered pricing** models and offering **discounts on long-term contracts** can boost customer retention.

4. Feature Correlation

A correlation analysis shows:

- **Contract type** has the strongest correlation (-0.40) with churn—long-term contracts **greatly reduce churn**.
- **Tech support and security services** negatively correlate (-0.28), meaning **offering these services improves retention**.
- **Higher charges** positively correlate ($+0.26$) with churn, showing **price-sensitive customers are more likely to leave**.

Recommendations:

- ✓ **Offer discounted annual plans** to retain month-to-month customers.
 - ✓ **Improve service experience** for fiber-optic customers.
 - ✓ **Encourage automatic payments** to reduce financial disruptions.
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Conclusion

This analysis confirms that **contract types, service satisfaction, pricing strategies, and payment methods** heavily influence customer churn. To **minimize churn**, businesses should implement **long-term engagement strategies, improve service quality, and adopt proactive customer support practices**.

By applying these strategies, companies can **significantly improve customer retention and revenue sustainability** over time.